

DIFFERENTIAL GROWTH

EVOLVING STRUCTURAL SYSTEMS FROM CLOSED GROWTH LOOPS

Cameron Hendey

ARC3201 - Brady Peters

2024

AIM

*TO CREATE DIFFERENTIAL GROWTH ALGORITHMS THAT CAN BE ADAPTED TO CREATE STRUCTURAL
SYSTEMS SUITABLE FOR 3D ASSEMBLY?*

CORE TERMS:

Parametric Design, Differential Growth, Structural Systems

DIFFERENTIAL GROWTH PATTERNS

DEFINITION:

The uneven expansion of a surface or curve over time, driven by *local differences in growth rates*. This phenomenon is observed in both natural systems and computational simulations.



NATURAL EXAMPLES



GIANT PUFFER FISH SKIN PATTERN



LETTUCE SEA SLUG BODY

RELEVANCE

- *Exploration of new structural systems for additive manufacturing that utilize one continuous line to create form*
- *Allows for structures to be built with one extruder or “head”*
- *Does not require the intense complete layering that typical 3D printing techniques use.*

SCOPE

INSIDE SCOPE:

- *Creating a code that generates connected structural systems through differential growth algorithms*

OUTSIDE SCOPE:

- *Exploring material properties or full-scale testing*

CONTEXT



ICD/ITKE RESEARCH PAVILION 2010



ICD/ITKE RESEARCH PAVILION 2015

STATE OF THE ART



THALLUS, ZAHA HADID

METHODS

WORKFLOW OVERVIEW:

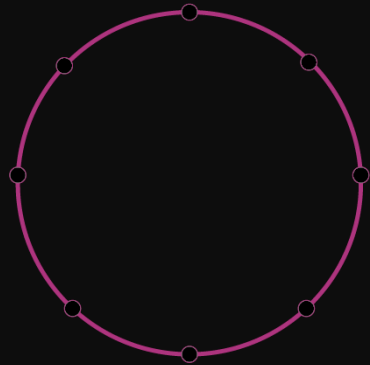
1. Explore basic differential growth code process
2. Experiment with usage and parameters
3. Debug and Fine tune code
4. Apply created code
5. Create applicable base surfaces to apply code to

TOOLS: RHINO, GRASSHOPPER, KANGAROO

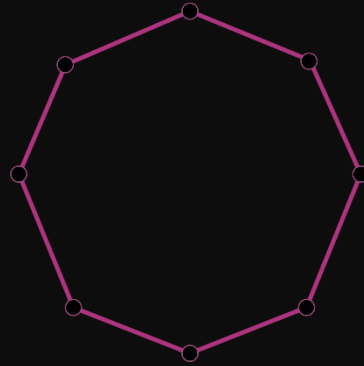
CODE DEVELOPMENT STEPS



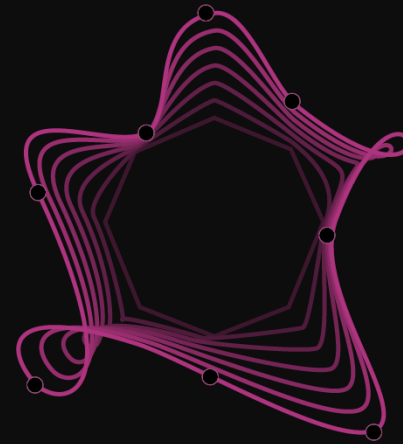
STEP 1:
DEFINE BASE CURVE



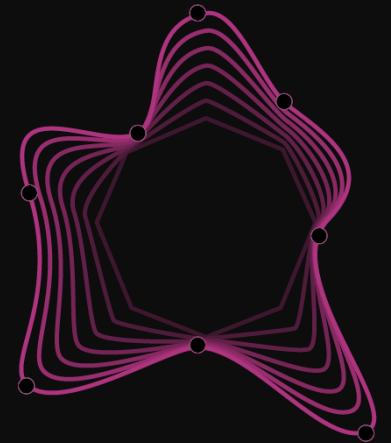
STEP 2:
SEGMENT CURVE



STEP 3:
DEFINE MOVEMENT &
GROWTH SEGMENTS



STEP 4:
BASE GROWTH ALGORITHM



STEP 2:
AVOID COLLISIONS &
OVERLAPS

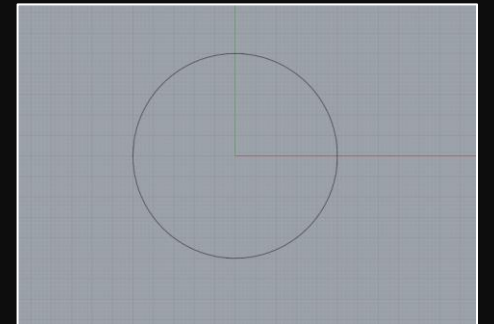
STEP 1: DEFINING BASE CURVE

REQUIREMENTS:

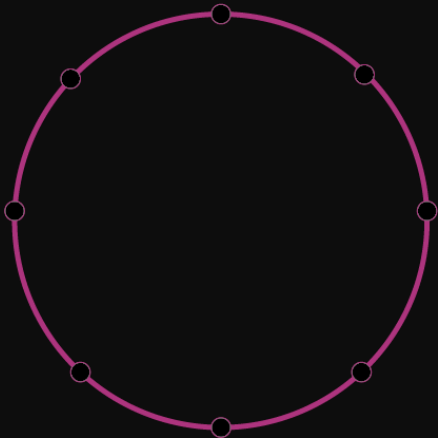
- Closed *planar* curve.
- Can be curved or have straight lines.

These shapes are used as the base curve before the code starts “growing” it.

Not super important but does impact how the growth patterns will form.

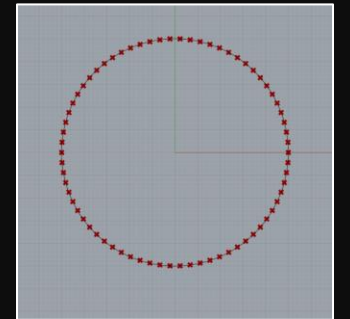
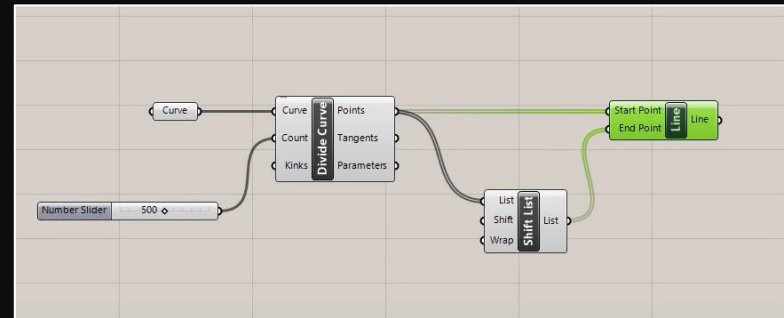


STEP 2: SEGMENTING CURVE

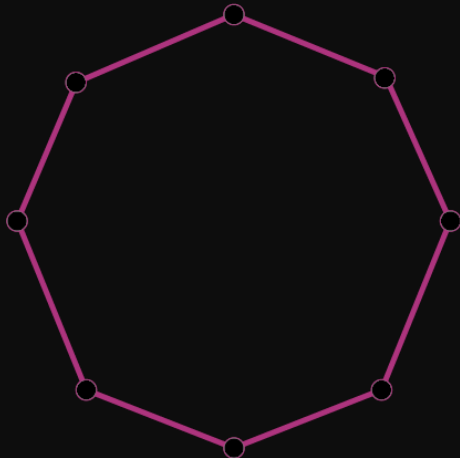
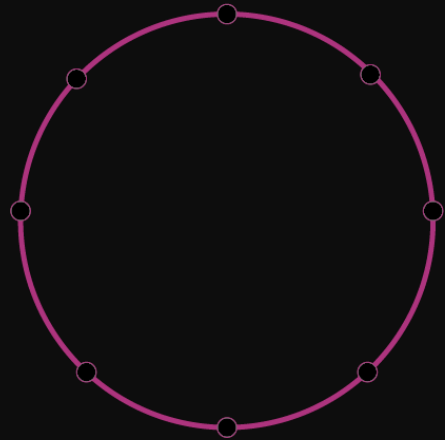


Divides the curve to create *nodes*

Nodes will move and start to *create growth patterns*



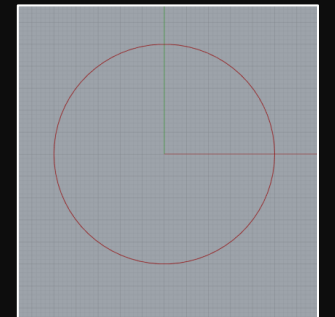
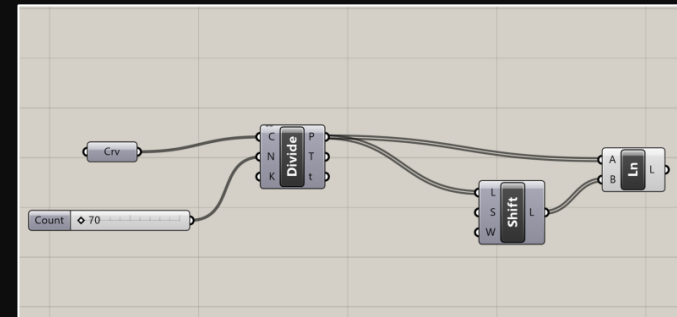
STEP 3: DEFINING NODES & SHAPE BUILDING



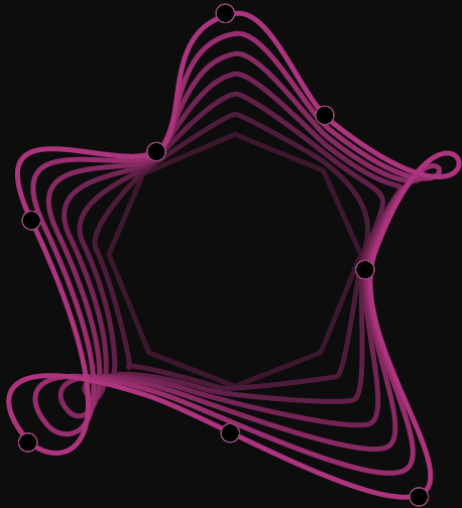
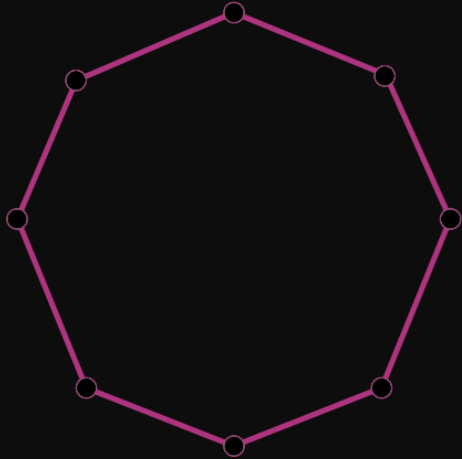
Creates *new lines* between each node

This will allow nodes to move while *retaining closed shape*

Allows (super) basic system of *pattern creation*



STEP 4: BASE GROWTH ALGORITHM

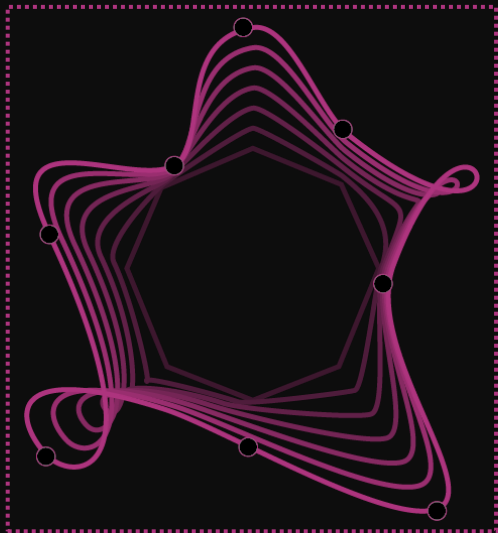
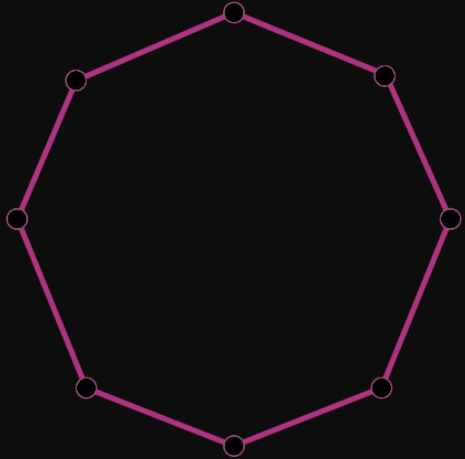


Goal is to make *each node start to move* in random directions

Retaining a *closed shape*

Implementation of *Kangaroo Physics*

STEP 4.a : CREATING & SETTING BOUNDARY

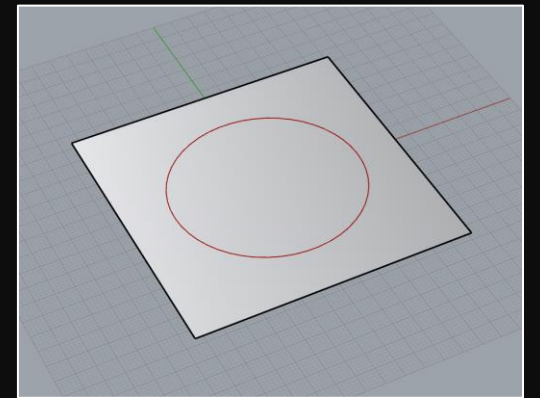


Create a bounding box for which the shape can *move* but is *limited* to

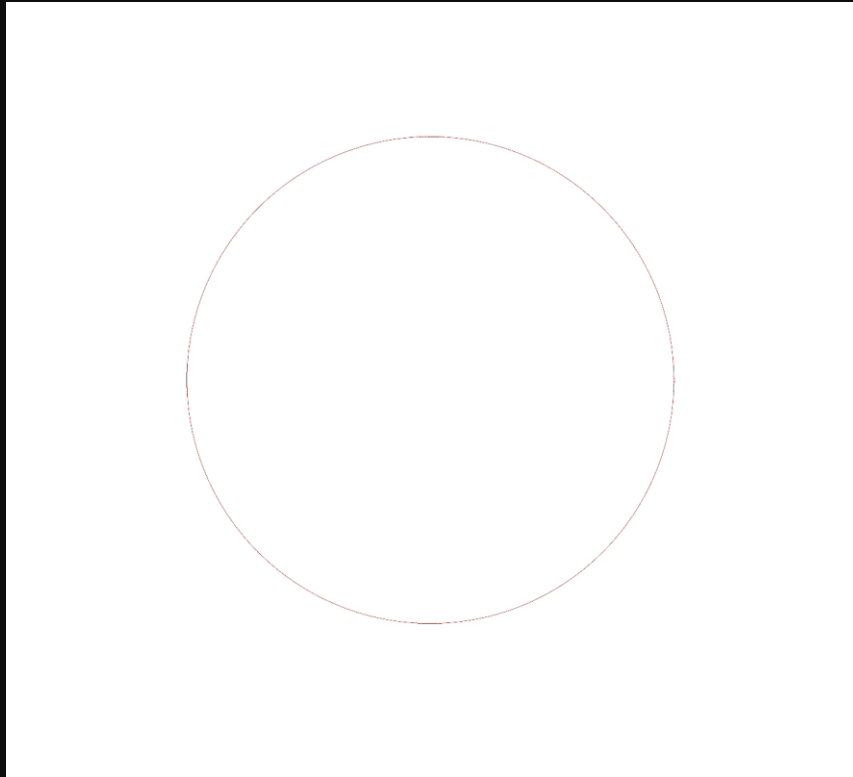
Began with a *simple planar form*

Goal is for this to become *3D and complex*

It will form the “*scaffolding*” of the final form



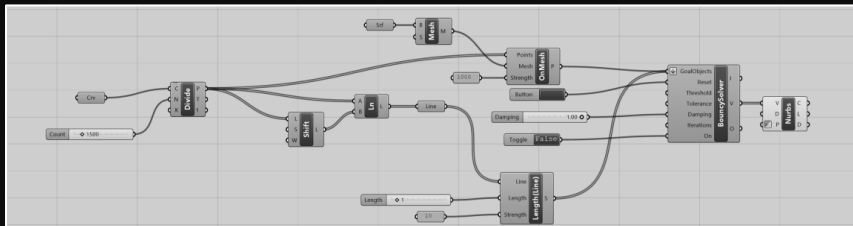
STEP 4.b : SIMPLE GROWTH PATTERN



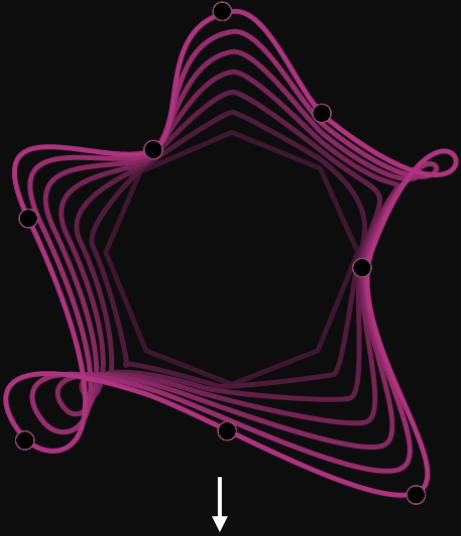
Using Kangaroo's *BouncySolver component* as the core

Feeding in the *nodes and base surface*

Giving lines that have momentum and growth but *overlap*

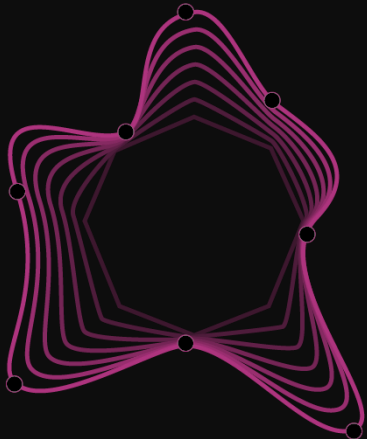


STEP 5: AVOIDING COLLISIONS & OVERLAPS

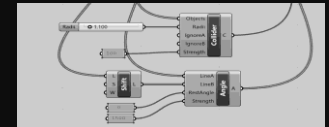
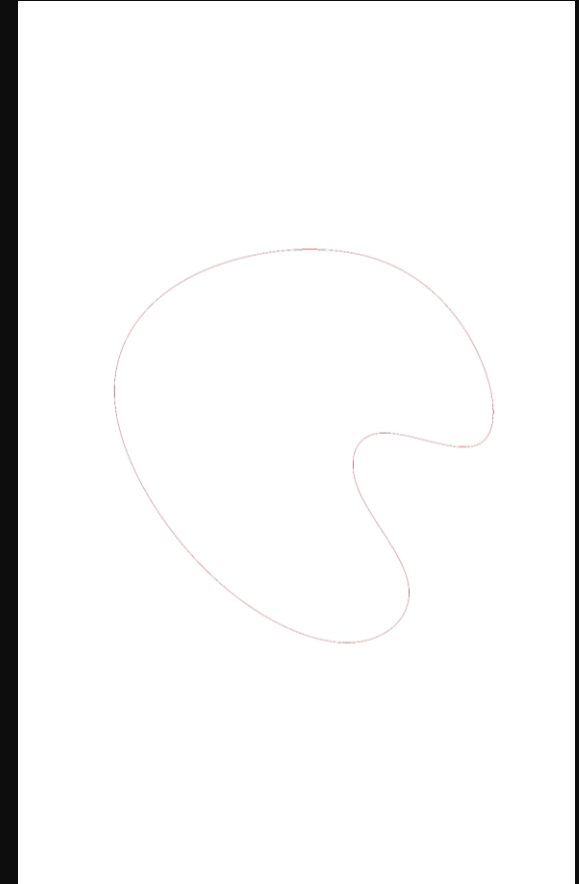


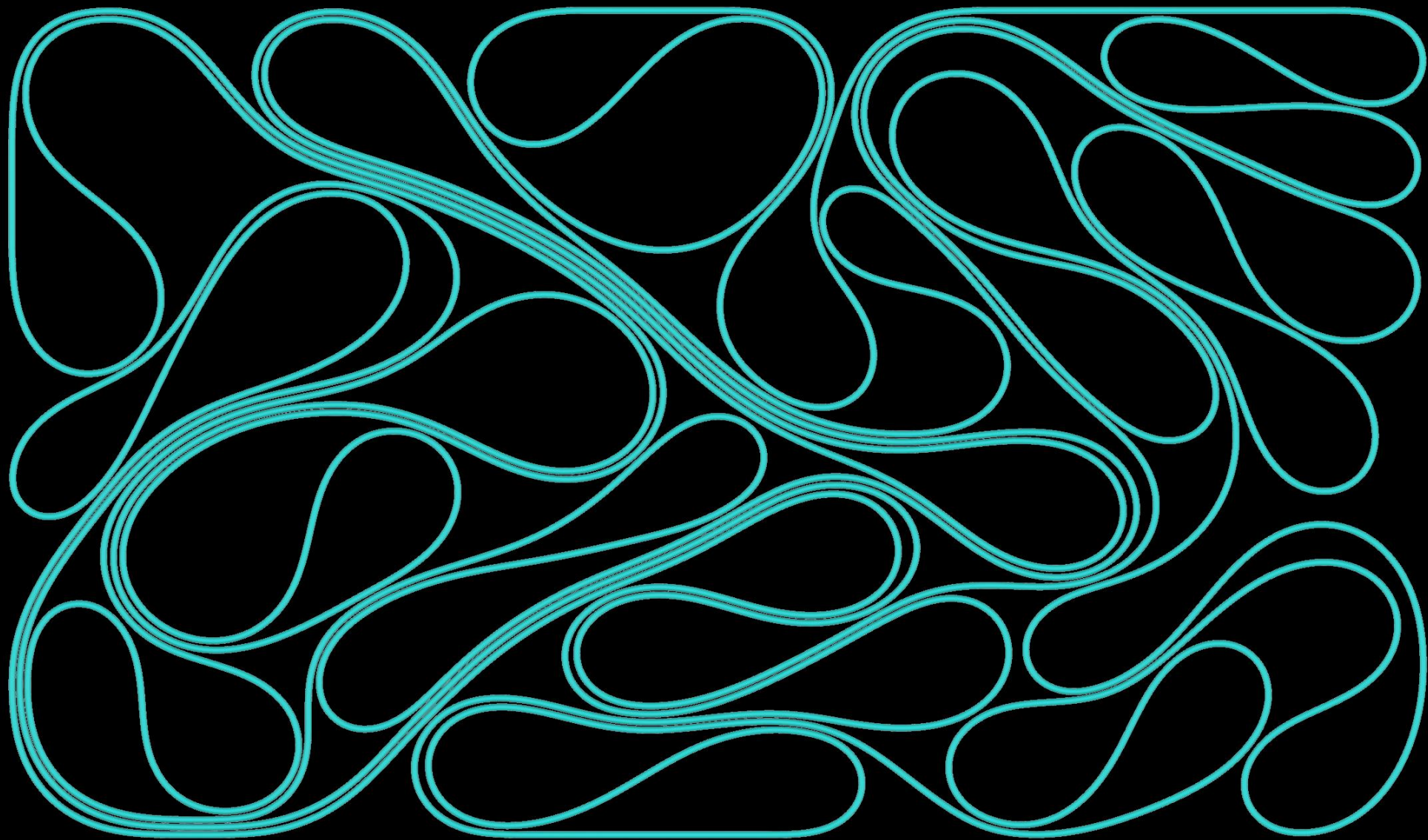
Used *colliders* to stop lines from intersecting

Started to create *iconic differential growth pattern*

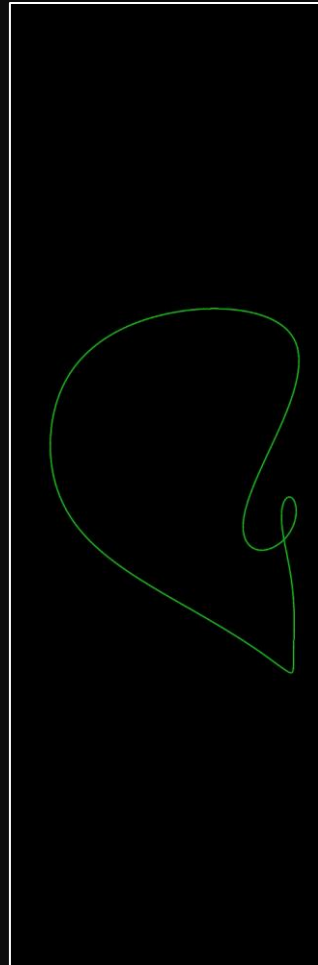
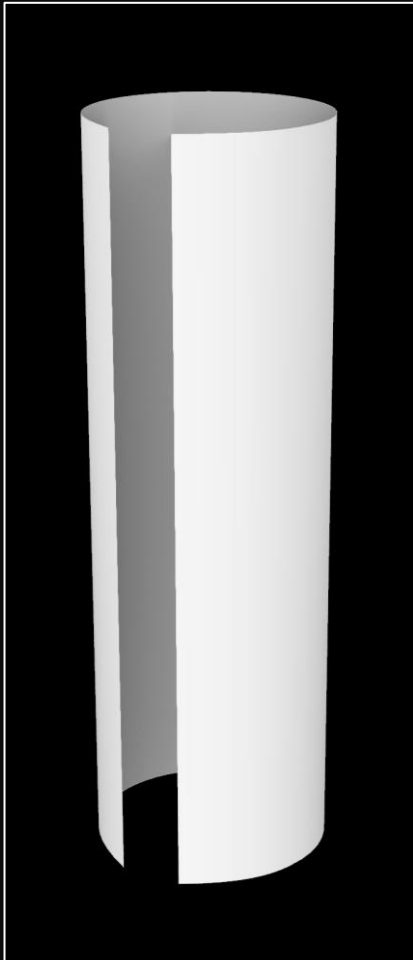


Added *angle limiters* to define curvature





STEP 6: 3D APPLICATION



Replace 2D bounding box with *3D form*

The growth pattern will take on this shape

It will create a form that can be created with *one*
extrusion line

ZAHA HADID'S THALLUS RECREATION

To recreate the effect on similar pieces can show the *program in work*

Thallus is one of the key inspiration for the project making it a key test of the program



REFERENCES

- Hadid, Zaha. *Thallus*. Milan's White in the City Exhibition, 2017.
Description of the project exploring innovative uses of differential growth patterns in computational design.
- Menges, Achim, and Jan Knippers. "ICD/ITKE Research Pavilion 2010: Integrating Design Computation and Robotic Fabrication." *Architectural Design*, vol. 80, no. 4, 2010, pp. 60-63.
Details the ICD/ITKE research pavilions and their influence on computational design and fabrication techniques.
- Menges, Achim. *Material Synthesis: Fusing the Physical and the Computational*. Springer, 2013.
Explores the integration of computational design and material properties, relevant to understanding parametric systems.
- "Differential Growth." *Generative Landscapes Blog*, generativelandscapes.wordpress.com/differential-growth/. Accessed 28 Nov. 2024.
A detailed explanation and examples of differential growth in both natural and computational contexts.
- Oxman, Neri. "Variable Property Rapid Prototyping: Inspired by Nature." *MIT Media Lab*, 2007.
Discusses innovative uses of computational design techniques inspired by natural growth patterns in fabrication.
- Piker, Daniel. *Kangaroo Physics Plugin for Grasshopper*. 2012.
Comprehensive documentation and examples of the Kangaroo Physics plugin, central to this project.
- Bouffanais, Roland. *Design and Control of Swarm Dynamics*. Springer, 2015.
Explains dynamic systems and algorithms like differential growth that influence computational design.
- Koch, Alexander J., and Hans Meinhardt. "Biological Pattern Formation: From Basic Mechanisms to Complex Structures." *Reviews of Modern Physics*, vol. 66, no. 4, 1994, pp. 1481-1507.
Explains biological pattern formation and its relevance to computational simulations.
- "Lettuce Sea Slug Biology." *Smithsonian Ocean*, ocean.si.edu/ocean-life/invertebrates/lettuce-sea-slug. Accessed 28 Nov. 2024.
Discusses the natural growth patterns of the Lettuce Sea Slug, a key inspiration for this project.
- Block, Philippe, and Timo Bächtold. *Digital Tectonics: Integrating Structural Engineering and Computational Design*. Wiley, 2018.
Explores the potential of computational design in creating innovative structural systems.

THANK YOU :)

